Same-root edge: independent source and composition review

Two actual find traces reach the same root
Derive endpoint affine equations and subtract both signed offsets

Opposite coefficients: constraint(-a,-b)
True edge excludes inconsistent zero coefficient; an answer equals k

Equal coefficients: invert +2 or -2, then anchor
Absent anchor is stored; repeated anchor compares old-minus-new coefficients

Global validity and post-find forest preserved
Independent controls include signed k offsets, both branches and invalid boundaries

6 main + 12 producer controls + 15 independent controls = 33 readbacks

Field/state model; native word/inverse/counter interfaces remain explicit